📘 MODULE 26: KNOWLEDGE OBJECTS

26.1 Concept

Knowledge objects are saved, reusable definitions that add meaning to your data and searches: saved searches, reports, alerts, macros, event types, tags, field extractions, calculated fields, field aliases, lookups, data models, workflow actions.

26.2 Why they matter professionally

They turn one analyst's knowledge into a **shared, consistent** team asset. This is the difference between an L1 who runs searches and an L2 who builds detections and content. Interviewers ask about them regularly.

26.3 The objects

Saved searches, reports, alerts

Covered in Module 28. All stored in `savedsearches.conf`.

Macros

Reusable SPL snippets. Settings → Advanced search → **Search macros** → New.

Without arguments

- Name: `auth_base`
- Definition: `index=linux_auth sourcetype=linux_secure`
- Use (note **backticks**, not quotes):

```spl
`auth_base` "Failed password" | stats count
```

With arguments

- Name: `failed_by_ip(1)` (the **number in parentheses is the argument count**)
- Arguments: `thr`
- Definition:

```spl
`auth_base` "Failed password" | rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})" | stats cou
```

- Use: `` `failed_by_ip(10)` ``

How it works: a macro is **text substitution** before the search runs. **Why it matters:** a change in one place updates every dashboard and alert that uses it. **Mistakes:** wrong arg count in the name, using quotes instead of backticks, and the macro not shared.

Event types

A **saved search that labels events**.

```ini
# eventtypes.conf
[failed_ssh_login]
search = sourcetype=linux_secure "Failed password"
```

Use: `eventtype=failed_ssh_login`. Eventtypes can't contain a pipe (`|`) or subsearch, which is a good interview fact. They are the bridge to **tags** and CIM.

Tags

A **label on a field value** (or event type).

```ini
# tags.conf
[eventtype=failed_ssh_login]
authentication = enabled
failure = enabled
```

Use: `tag=authentication`. Tags power CIM data model constraints.

Field extractions

`EXTRACT-` and `REPORT-` in `props.conf` (Module 23), or built with the Field Extractor.

Calculated fields

`EVAL-<field> = <expression>` in `props.conf`. Computes a field automatically at search time, for example `EVAL-action = if(status=="success","success","failure")`. **Limitation:** each is evaluated on events of that sourcetype only.

Field aliases

`FIELDALIAS-<name> = <old> AS <new>`. Gives an existing field an extra name (for example `src_ip AS src`). The original field is **kept**.

Lookups

Module 11. Includes automatic lookups (`LOOKUP-`).

Workflow actions

Clickable links or searches attached to a field, shown in the event's actions menu. Settings → Fields → Workflow actions → New.

- Type: **Link**. Example: URL `https://www.virustotal.com/gui/ip-address/$src_ip$`, applied to field `src_ip`. Clicking it opens an external lookup for that IP value.
- Type: **Search**. Example: open a new search with `index=linux_auth src_ip=$src_ip$`.
- **SOC use:** one-click IP/domain/hash enrichment.

Data models

Module 16.

26.4 Where they live and how they're shared

- Each object belongs to an **app** and an **owner**, with Private/App/Global permissions (Module 25.7).
- Stored in that app's `local/` folder (`macros.conf`, `eventtypes.conf`, `tags.conf`, `savedsearches.conf`, `workflow_actions.conf`).
- **Orphaned objects:** if a user who owns scheduled searches is deleted, those searches can become orphaned and stop running. Admins check for them with **Settings → All configurations** (filter by orphaned) or reassign knowledge objects. It's a real admin interview topic.

26.5 Naming and governance (professional habits)

- Naming pattern: `<team>_<purpose>_<type>` (for example `soc_bruteforce_alert`).
- Keep macros small and documented, and reuse them.
- Share to **App**, not Global, unless needed.
- Document owner and purpose in the description field.

26.6 Hands-on lab

1. Create macro `auth_base` and use it in a search.
2. Create macro `failed_by_ip(1)` and test `` `failed_by_ip(3)` ``.
3. Create event type `failed_ssh_login`, then tag it `authentication`, then run `tag=authentication`.
4. Add a field alias `src_ip AS src` for `linux_secure` (Settings → Fields → Field aliases) and confirm `src` appears.
5. Create a calculated field `action` for `linux_secure`.
6. Create a workflow action on `src_ip`.
7. Share them all with **App** scope, then check that a second user sees them.

26.7 Troubleshooting

Symptom	Cause
Macro "not found"	Wrong name or arg count, not shared, or wrong app
Eventtype returns nothing	Search text doesn't match, or it contains a pipe
Tag not appearing	Tag set on a different eventtype/field value, or not shared
Alias or calculated field missing	Sourcetype mismatch, wrong app/permission
Scheduled search stopped after an employee left	Orphaned object, so reassign ownership

26.8 Interview Q&A

- **What are knowledge objects?** Reusable, shareable definitions that add structure and meaning to data and searches.
- **What is a macro and how do you call it?** Reusable SPL text, called with backticks, optionally with arguments.
- **Event type vs tag?** An event type is a saved search that labels events; a tag is a label on a field value or event type, used to group them.
- **Alias vs calculated field?** An alias gives a second name; a calculated field computes a new value with `eval`.
- **Why can't others see my knowledge object?** Private scope.
- **What is an orphaned search?** A scheduled object whose owner no longer exists. It stops running until reassigned.
- **How do knowledge objects relate to CIM?** Aliases, calculated fields, event types, and tags normalize data so it appears in the data models.

26.9 Revision

Macros use backticks and `(n)` args · event type = labeled search (no pipes) · tag = label on a value · alias keeps original · calculated field = `EVAL-` · workflow action = click-through · permissions Private/App/Global · watch orphans.

✅ Self-Check (no peeking)

1. Name the two config files on a UF and what each does.
2. Why must filtering/masking config go on the HF if one is in the path?
3. What do `SOURCE_KEY`, `DEST_KEY`, and `FORMAT` mean?
4. What does `nullQueue` do, and does the event count against the license?
5. HEC returns `Success` but your search shows nothing. Name three causes.
6. Streaming vs transforming command: give two examples each.
7. Why is a macro called with backticks, and what does `foo(2)` mean in its name?

🧪 Exercises (write them yourself)

1. Write `props.conf` and `transforms.conf` to **drop every event containing** `healthcheck` for sourcetype `web_app`.
2. Write a `SEDCMD` that masks credit-card-like 16-digit numbers.
3. Create the `curl` command to send a JSON event with `user=bob` to HEC (token placeholder), into index `main`.
4. Write a macro with one argument that returns failed logins for a given `host`, and show how you'd call it.
5. A UF shows **Active** in `list forward-server`, but `index=linux_auth` is empty. List your first four checks.